Exercise 1: Investigation Time

```
#include <iostream>
using namespace std;

void analyzeInvestigation(int roomCount, int &sumTime, int &longRoomCount)
{
    int timeSpent;
    sumTime = 0;
    longRoomCount = 0;

    for (int i = 1; i <= roomCount; i++)
    {
        cout << "Enter minutes spent in room " << i << ": ";
        cin >> timeSpent;

        sumTime += timeSpent;
        if (timeSpent > 20)
            longRoomCount++;
    }
}

int main()
{
    int roomCount;
    int sumTime;
    int longRoomCount;

    cout << "Enter number of rooms: ";
    cin >> roomCount;

    analyzeInvestigation(roomCount, sumTime, longRoomCount);

    cout << endl;
    cout << "Total investigation time: " << sumTime << " minutes" << endl;
    cout << "Number of challenging rooms: " << longRoomCount << endl;

    return 0;
}
```

Exercise 2: Minutes Until Midnight

```
#include <iostream>
using namespace std;

int minutesUntilMidnight(int minutesPassed)
{
    return 1440 - minutesPassed;
}

int minutesUntilMidnight(int clockHour, int clockMinute)
{
    int minutesPassed = clockHour * 60 + clockMinute;
    return minutesUntilMidnight(minutesPassed);
}

int main()
{
    int clockHour;
    int clockMinute;
    int minutesPassed;

    cout << "Enter hour and minute: ";
    cin >> clockHour >> clockMinute;
    cout << "Minutes until midnight: " << minutesUntilMidnight(clockHour, clockMinute) << endl;

    cout << "Enter elapsed minutes for the second clock: ";
    cin >> minutesPassed;
    cout << "Minutes until midnight: " << minutesUntilMidnight(minutesPassed) << endl;

    return 0;
}
```

Exercise 3: Room Investigation

```
#include <iostream>
using namespace std;

void investigate(int &currentEnergy, int &totalClues, int roomLevel)
{
    int energyCost = 2 * roomLevel;

    if (currentEnergy >= energyCost)
    {
        currentEnergy -= energyCost;
        totalClues += roomLevel;
        cout << "Investigation completed" << endl;
    }
    else
    {
        cout << "Not enough energy" << endl;
    }
}

void investigate(int &currentEnergy, int &totalClues, int roomLevel, int hintValue)
{
    int energyCost = 2 * roomLevel - hintValue;

    if (currentEnergy >= energyCost)
    {
        currentEnergy -= energyCost;
        totalClues += roomLevel;
        cout << "Investigation completed" << endl;
    }
    else
    {
        cout << "Not enough energy" << endl;
    }
}

int main()
{
    int currentEnergy;
    int roomTotal;
    int totalClues = 0;
    int roomLevel;
    int hasHelp;
    int hintValue;

    cout << "Enter initial energy: ";
    cin >> currentEnergy;
    cout << "Enter number of rooms: ";
    cin >> roomTotal;

    for (int r = 1; r <= roomTotal; r++)
    {
        cout << "Room " << r << " - enter difficulty and help (1/0): ";
        cin >> roomLevel >> hasHelp;

        if (hasHelp == 1)
        {
            cout << "Enter hints: ";
```

```
            cin >> hintValue;
            investigate(currentEnergy, totalClues, roomLevel, hintValue);
        }
        else
        {
            investigate(currentEnergy, totalClues, roomLevel);
        }

        cout << "Energy: " << currentEnergy << " Clues: " << totalClues << endl;
    }

    return 0;
}
```